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Principal Data Engineer and Architect – Data Engineering & Cloud

Principal Data Engineer and Architect with 15+ years of expertise in designing and building petabyte-scale, cloud-native data platforms for global enterprises. Deep hands-on technical proficiency in distributed systems (Spark, Kafka), multi-cloud architectures (AWS, Azure, GCP), and full-stack data engineering, from real-time ingestion to optimized storage and governance. Proven ability to set technical strategy, define reference architectures, and solve complex data problems at scale, driving innovation and significant cost efficiency.

PROFILE SUMMARY

- **Strategic Leadership in Data Platform Architecture:** Expert in designing and executing data strategy, building petabyte-scale data platforms from the ground up, and modernizing legacy systems to cloud-native architectures (AWS, Azure, GCP) to serve as the foundational bedrock for product and analytics.
- **Full-Lifecycle Cloud Data Transformation:** Led multiple large-scale migrations of on-premise Hadoop ecosystems to modern cloud data stacks (AWS Glue/Lambda/S3, Azure Data Factory/ADLS/Synapse), achieving 40%+ reduction in TCO while significantly improving scalability, reliability, and time-to-insight.
- **Engineering Excellence & DevOps Culture:** Champion of DevSecOps and CI/CD for data platforms. Instituted automated testing, deployment pipelines (Jenkins, GitLab, Ansible), and infrastructure-as-code practices, slashing deployment times by 70% and enabling rapid, reliable iteration for data teams.
- **Real-Time Data Ecosystem Development:** Architected and managed low-latency, high-throughput data ingestion frameworks using Kafka, Spark Streaming, and Flink to power real-time feature computation, personalization engines, and live dashboards for end-user products.
- **Data Governance & Reliability Engineering:** Established frameworks for data quality, lineage, and metadata management (e.g., DataHub, Great Expectations), increasing data trustworthiness and reducing data-related incidents by over 50% for downstream consumers.
- **Solution Architecture & Stakeholder Alignment:** Translates complex business and product requirements into robust technical specifications and architectural blueprints, ensuring data solutions are scalable, reliable, and aligned with long-term strategic goals.
- **Technical Mentorship & Best Practices:** Acts as a technical mentor and subject matter expert, elevating the skills of engineers through code reviews, architectural guidance, and establishing best practices for data modeling, coding, and system design.
- **Cost-Efficient & Performance-Optimized Systems:** Deeply focused on the business impact of technology. Expert in performance tuning of distributed systems (Spark, Hive) and implementing cost-optimization strategies for cloud data platforms, resulting in annual savings of hundreds of thousands of dollars.
- **ML & AI Data Infrastructure:** Hands-on experience in building the data backbone for AI/ML initiatives, including feature store development, orchestration of large-scale data processing for model training, and supporting MLOps workflows to operationalize models efficiently.
- **Data-Driven Decision Making:** Leverages strong analytical skills to define and monitor KPIs and SLOs for data platforms. Uses metrics on data quality, pipeline performance, and system health to drive strategic decisions, prioritize resources, and demonstrate clear ROI on data investments.

EXCELLENCE SPHERE

Enterprise Data Architecture || AI & ML Integration Strategy || Data Ingestion & Integration Frameworks || Cloud Transformation

Big Data Ecosystems || Project & Program Management || Generative AI Implementation || Data Governance & Compliance

Platform Modernization || DevSecOps& CI/CD Automation || MLOps || Model Lifecycle Management

Agile Delivery &SAFe Frameworks || Infrastructure & Application Migration || Digital Transformation Leadership

AI-driven Analytics || Prompt Engineering || Regulatory & Risk Compliance || Data Modeling || Distributed Systems || Performance Engineering || Cost Optimization || POC Development || Technology Evaluation || Standards & Best Practices || Operational Resilience & Root Cause Analysis || Quality Engineering || KPI-Driven Delivery Execution

TECHNICAL SKILLS

Big Data Ecosystem:	Hadoop (HDFS, Hive, Pig, HBase, Sqoop, ZooKeeper, Flume, Oozie), Spark (2.x, 3.x), NiFi, Kafka, HDP, CDP
Programming Languages:	Python, Scala, Core Java, R, Shell Scripting
Cloud Services (AWS):	S3, AWS Glue, Lambda, Athena, Kinesis, DynamoDB, SageMaker
Cloud Services (Azure):	Azure DevOps (ADO), Data Factory, ADLS Gen2, Data Lake, Delta Lake, Synapse Analytics, ML Studio
Cloud Services (GCP):	BigQuery, Vertex AI, Cloud Storage, Dataflow
ML/AI Frameworks & Tools:	TensorFlow, PyTorch, Hugging Face, LangChain
Vector Databases:	Pinecone, Weaviate, FAISS
Databases:	Oracle, Greenplum, PostgreSQL, MongoDB, HBase
BI & Visualization:	Tableau, Splunk
Other Tools:	IBM CDC, Talend

PROFESSIONAL CERTIFICATIONS & TRAINING

PMP®, Project Management Institute
Data Landscape of GenAI for Project Managers from PMI®
Practical Application of Generative AI for Project Managers from PMI®
Cloudera Certified Developer for Apache Hadoop (CCDH)Apache Hadoop
AWS Training - Genpact, 2018 (2-week period)
MS BI (SSIS, SSAS, SSRS) Corporate Training - Tech Mahindra, 2011 (7-day training)
Hadoop Corporate Training - Tech Mahindra, 2011 (15-day training)

PROFESSIONAL EXPERIENCE

Ingestion Framework Lead – Assistant Vice President, Standard Chartered Bank – GBS, Chennai, India (Apr 2022 – Present)

Key Deliverables:

- Architected a multi-cloud capable enterprise ingestion framework handling **>5 TB of data daily** from **200+ sources** (Oracle, DB2, SQL Server, flat files). Designed the framework core to be cloud-agnostic, with **Azure (ADF, ADLS Gen2, Delta Lake)** as the primary deployment, while ensuring portability to **AWS (S3, Glue)** and **GCP (Cloud Storage, Dataflow)**.
- Engineered the framework to leverage **IBM CDC** for real-time data capture and **Apache NiFi & Spark** for orchestration and transformation, enabling seamless batch and real-time processing into the cloud data lake.
- Pioneered the implementation of a unified **DevSecOps CI/CD pipeline** using **Azure DevOps (ADO)** and GitLab to automate the deployment of data pipelines across environments, **improving deployment frequency by 80%** and ensuring security and compliance checks were embedded in the workflow.
- Developed and enforced data quality and governance checks within the ingestion framework, **improving data reliability for downstream consumers by 60%** and ensuring adherence to enterprise standards.
- Partnered with the Cyber security and Enterprise Architecture teams to embed security controls and data governance policies directly into the ingestion framework, achieving and maintaining compliance with **GDPR and BCBS 239** regulations.
- Authored and socialized KPI dashboards for platform health (uptime, latency, data quality scores), creating transparency and enabling data-driven decision-making for leadership on resource allocation and strategic priorities.
- Championed cross-functional collaboration across Technology, Product, and Business stakeholders to align platform capabilities with evolving digital banking requirements, improving time-to-market and operational synergy.

Technical Lead – Big Data Platform, Uber India Research and Development, Bengaluru, India (Sep 2020 – Apr 2022)

Key Deliverables:

- Led the development and optimization of large-scale ETL pipelines on Uber's **on-premise Hadoop (HDFS, Hive)** and **Spark** ecosystem, processing **petabyte-scale** raw and core data tables to serve critical business intelligence and machine learning models.
- Optimized complex **Apache Spark (Scala/PySpark)** and **Hive** jobs, **reducing average ETL job runtime by 35%** and improving the efficiency of the entire data processing platform.
- Collaborated with data platform engineers to enhance framework capabilities for data lineage and metadata management, increasing transparency and trust in data assets across engineering teams.
- **Served as the technical lead and primary architect** for the data platform team, providing expert guidance on Spark optimization, data modeling, and pipeline design to ensure the timely delivery of quarterly objectives.

Lead Consultant - Genpact, Hyderabad, India (Sep 2015 – Dec 2020)

Key Deliverables:

- Architected and implemented a scalable **on-premise Hadoop (HDP)** data processing framework for ingesting and analyzing **2+ TB of machine logs daily**, using **Spark Streaming** and **Kafka** to enable real-time operational analytics.
- Led the design and execution of a cloud modernization initiative, migrating complex on-premise data pipelines to **AWS**. Re-engineered workflows using **AWS Glue (PySpark), Lambda, and S3**, designing a serverless architecture that **reduced TCO by 40%**.
- Developed reusable frameworks and templates for data ingestion and transformation in **AWS**, significantly accelerating the development of new data pipelines and standardizing best practices across projects.
- Built robust, scalable data pipelines from various RDBMS and streaming sources into the data lake, ensuring data was available and optimized for analytical consumption.

Senior Data Engineer - L&T Technology Services (Apr 2014 – Sep 2015) ~ Tech Lead-Tech Mahindra Ltd (Dec 2005 – Apr 2014)

Key Deliverables:

- Laid the foundation for big data expertise by developing and optimizing large-scale **ETL processes** using **Hive, Pig, and custom scripts** to process TBs of data within early Hadoop ecosystems.
- Designed and implemented data integration pipelines and data warehouses, ensuring high data quality and availability for business intelligence and reporting functions.
- Gained deep foundational knowledge in database management, performance tuning, and system architecture, which informed later cloud and distributed systems designs.

EDUCATION

- **Master of Computer Applications (M.C.A.)**, Madras University, Chennai, India (2003)
- **Bachelor of Science in Computer Science**, Kakatiya University, Warangal, India (2000)